VMware vSphere

Infrastructure Implementation Project

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1.Introduction:

VMware vSphere is a virtualization platform that allows multiple virtual machines to run on physical servers. It helps organizations reduce hardware costs, improve efficiency, and manage IT resources more effectively.

vSphere provides tools for creating, managing, and monitoring virtual machines in a data center environment. The platform includes ESXi (the hypervisor) and vCenter Server (the management tool).

This project aims to design and implement a VMware vSphere environment by deploying multiple ESXI hosts and a centralized vCenter Server. The setup demonstrates core virtualization concepts such as clustering, shared storage, networking, and advanced features like vMotion, High Availability (HA), Distributed Resource Scheduler (DRS), and Fault Tolerance (FT). The environment provides a practical lab for managing virtual machines and validating enterprise level virtualization capabilities.

2.**Overview**

- **ESXi Hosts**
  - **Three ESXi servers: ESXi-0, ESXi-1, ESXi-2**
  - **ESXi-0 also hosts the vCenter Server Appliance**

- **vCenter Server**
  - **Deployed on ESXi-0**
  - **Provides centralized management for all ESXi hosts**
  - **Used to configure and monitor the cluster**

- **Cluster Configuration**
  - **ESXi-0, ESXi-1, and ESXi-2 combined into a single cluster**
  - **Advanced features enabled: vMotion, HA, DRS, FT**

- **Networking**
  - **vSwitch0: Management + VM traffic**
  - **vSwitch1: Storage + vMotion traffic**

- **Storage**
  - **Shared NFS datastore accessible by all ESXi hosts**
  - **Required for HA, DRS, and FT functionality**

- **Virtual Machines**
  - **Created and managed via vCenter**
  - **Operations include cloning, snapshots, and templates**
  - **Migration tested across ESXi hosts**

3.Installations:

During the installation process, we deployed three ESXi hosts (ESXi-0, ESXi-1, and ESXi-2) as virtual machines. Each host was configured with a **static IP address**, which we considered essential for stable connectivity and reliable integration with vCenter. Static addressing ensured that advanced features such as vMotion, High Availability (HA), and Distributed Resource Scheduler (DRS) could operate without interruption.

The **vCenter Server Appliance (VCSA)** was installed on ESXi-0. We carefully assigned static IP, DNS, and gateway during setup to maintain consistent communication with the ESXi hosts. Resource allocation was planned to provide sufficient CPU and memory for smooth operation, and we verified DNS resolution and time synchronization, as these are critical for host registration and cluster services.

3.1Network Configuration Summary:

The following tables detail the IP addressing scheme and network configuration for all infrastructure components**.**

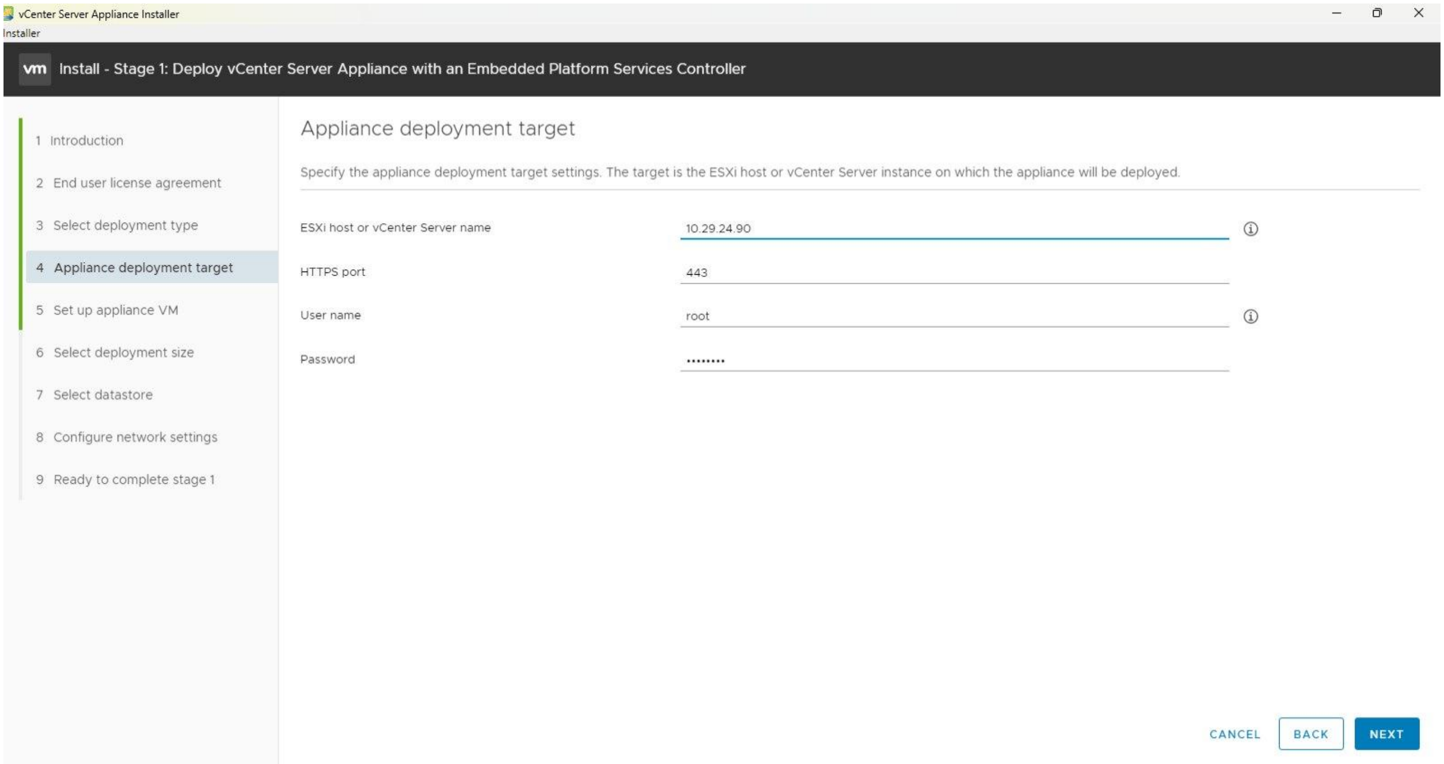
Network Parameters:

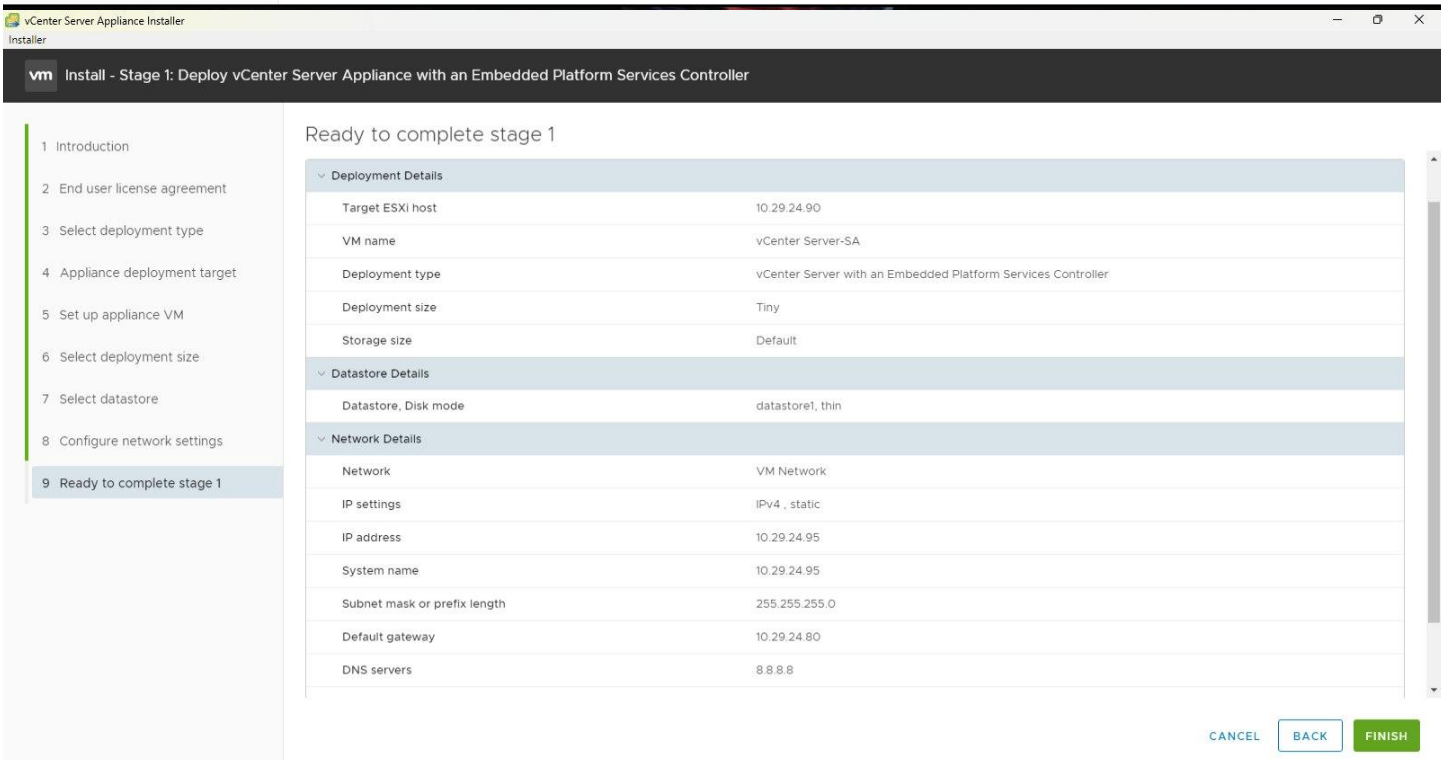
Parameter	**Value**
Network Address	**10.29.24.0**
Subnet Mask	**255.255.255.0**
Gateway	**10.29.24.80**

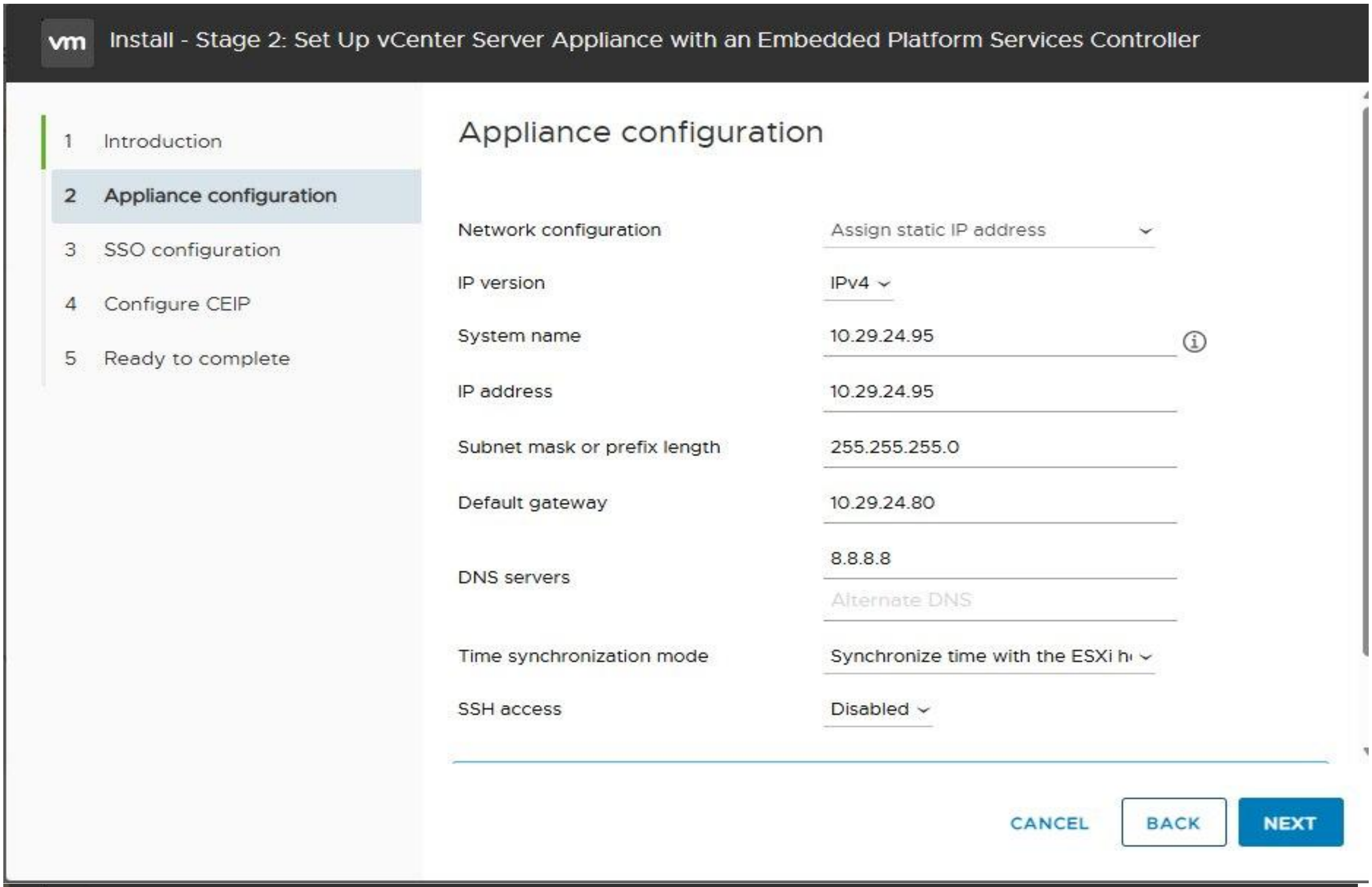
IP Address Assignment:

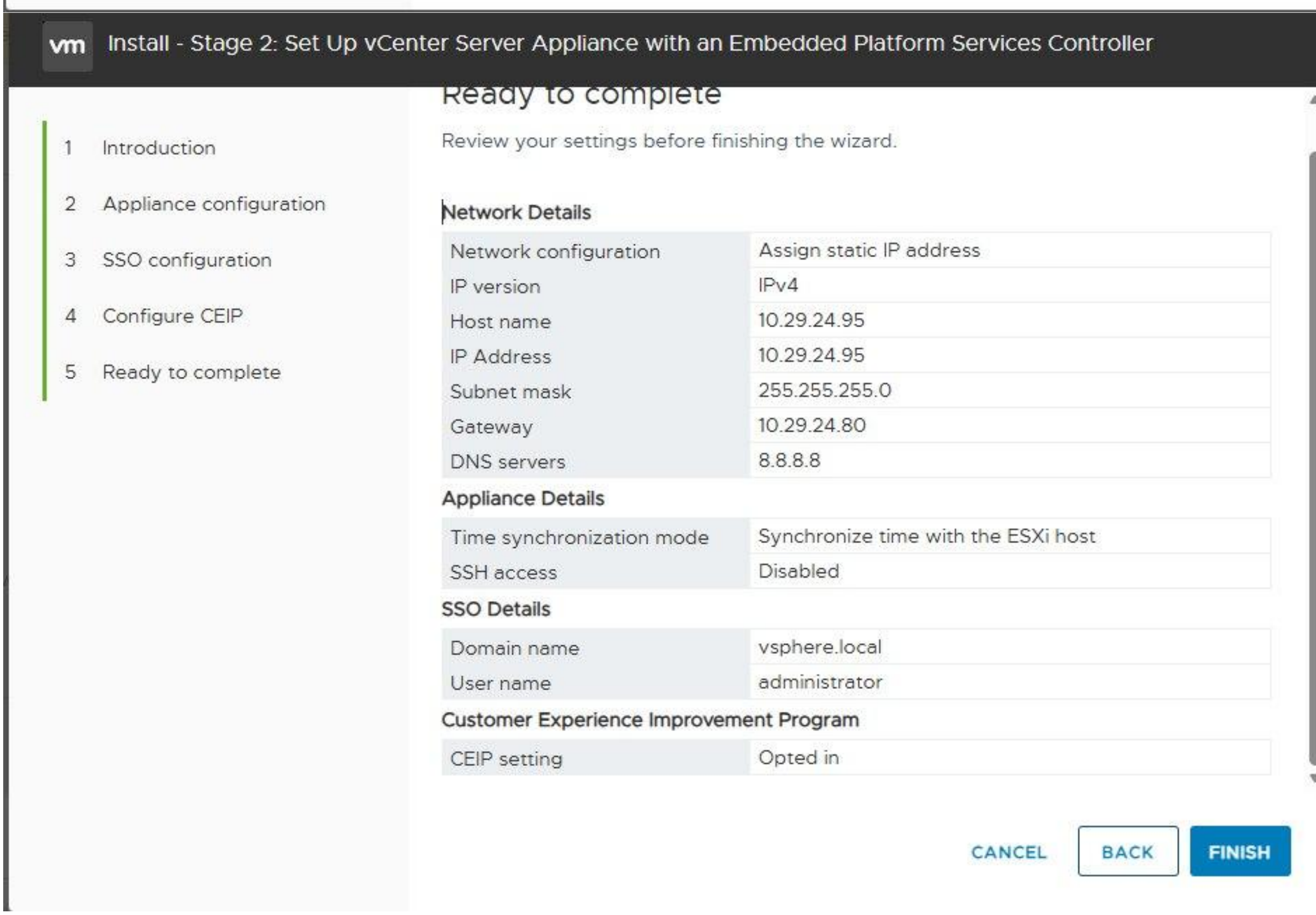
Device	**IP Address**	**Role/Description**
ESXi-00	**10.29.24.90**	**ESXi Host_00**
ESXi-01	**10.29.24.91**	**ESXi Host_01**
ESXi-02	**10.29.24.92**	**ESXi Host_02**
vCenter Server	**10.29.24.95**	**vCenter Server (Centralized Management)**
NFS-Storage	**10.29.24.25**	**Linux NFS Shared Storage Server**

VCenter deployment:
Images of installation:

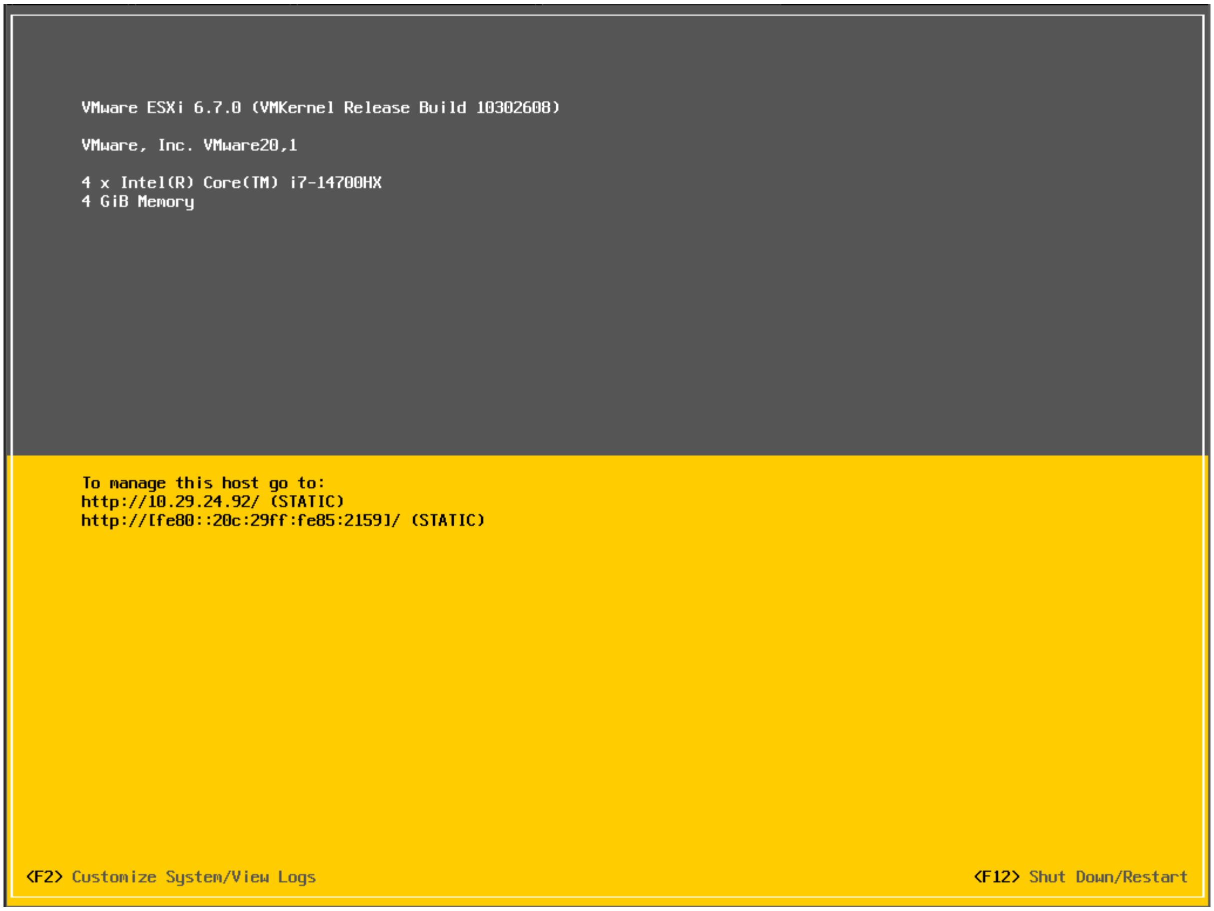


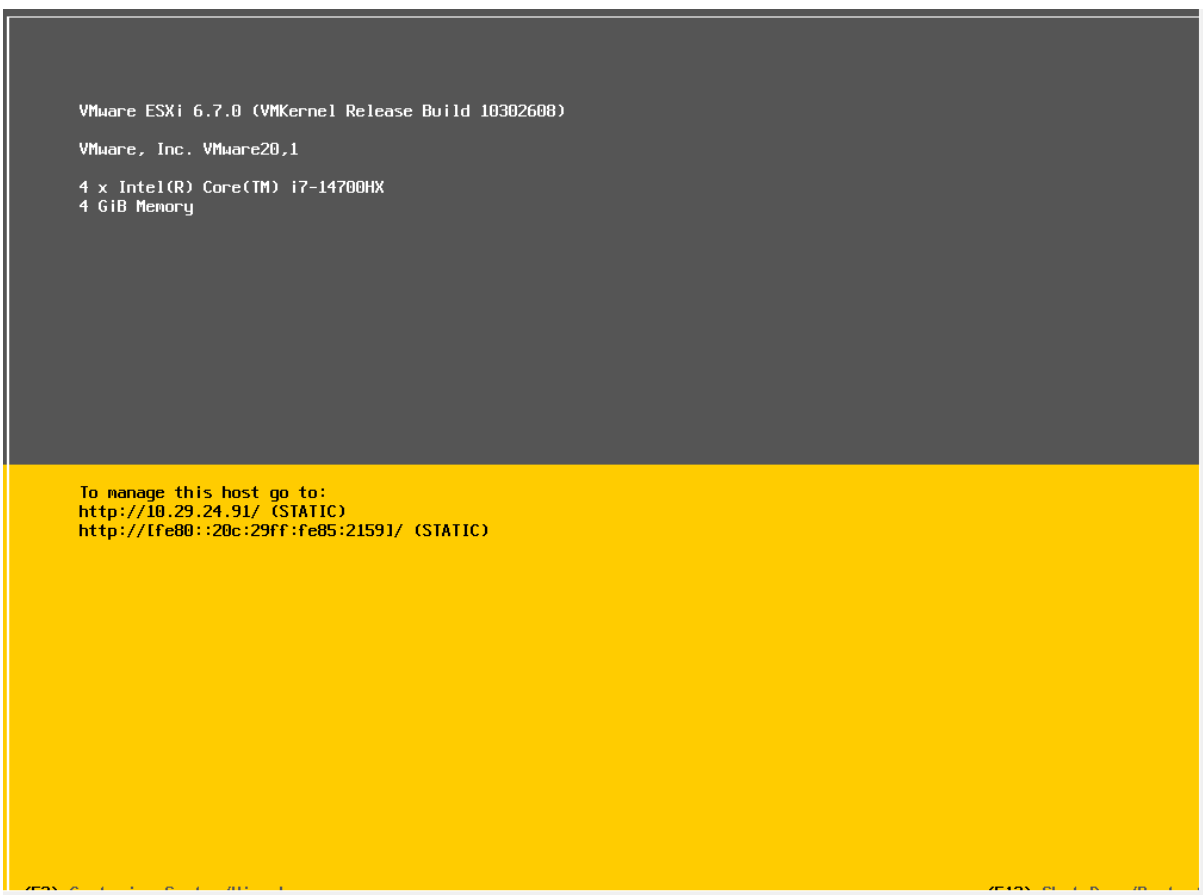



vm
Install - Stage 2: Set Up vCenter Server Appliance with an Embedded Platform Services Controller
1 Introduction
2 Appliance configuration
3 SSO configuration
4 Configure CEIP
5 Ready to complete
Appliance configuration
Network configuration
Assign static IP address
IP version
IPv4
System name
10.29.24.95
IP address
10.29.24.95
Subnet mask or prefix length
255.255.255.0
Default gateway
10.29.24.80
DNS servers
8.8.8.8
Alternate DNS
Time synchronization mode
Synchronize time with the ESXi h
SSH access
Disabled
CANCEL
BACK
NEXT


vm
Install - Stage 2: Set Up vCenter Server Appliance with an Embedded Platform Services Controller
Ready to complete
1 Introduction
2 Appliance configuration
3 SSO configuration
4 Configure CEIP
5 Ready to complete
Review your settings before finishing the wizard.
Network Details
Network configuration
Assign static IP address
IP version
IPv4
Host name
10.29.24.95
IP Address
10.29.24.95
Subnet mask
255.255.255.0
Gateway
10.29.24.80
DNS servers
8.8.8.8
Appliance Details
Time synchronization mode
Synchronize time with the ESXi host
SSH access
Disabled
SSO Details
Domain name
vsphere.local
User name
administrator
Customer Experience Improvement Program
CEIP setting
Opted in
CANCEL
BACK
FINISH

ESXi-1 and ESXi-2:



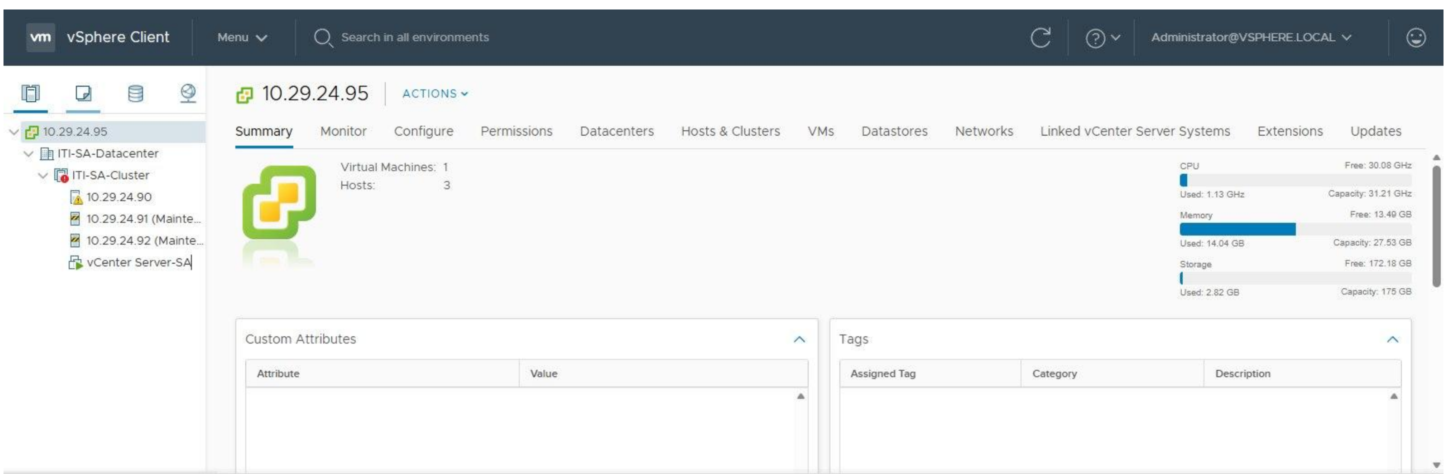


4. vCenter Cluster Configuration:

Cluster configuration in VMware vSphere involves grouping multiple ESXi hosts under centralized management in vCenter to create a single logical unit. This setup allows resources such as CPU, memory, and storage to be pooled together, enabling advanced features like vMotion, High Availability (HA), Distributed Resource Scheduler (DRS), and Fault Tolerance (FT).

To join ESXI hosts to a cluster, each host is first added to vCenter using its management IP and root credentials. Once connected, the hosts are placed into the cluster, where they share access to the same datastore and networking configuration. This ensures that virtual machines can be migrated between hosts, workloads can be balanced automatically, and services remain available even if one host fails. In essence, cluster configuration transforms individual ESXi servers into a resilient and flexible environment for enterprise-level virtualization.

- Connecting the ESXi hosts to vCenter Server and combining them into a cluster:



- Enabling the HA and DRS:

5. Networking Setup (vSwitches)

To properly separate traffic types, networking was configured directly on each ESXI host.
First, an additional Network was added to each Vswitch on each host to provide dedicated bandwidth for storage and vMotion traffic. Using the ESXI host client.

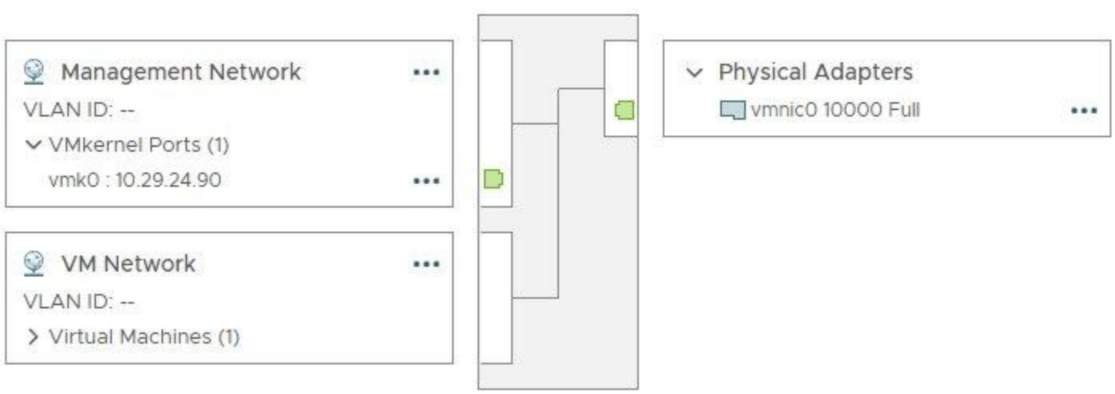
• On vSwitch1, two port groups were defined:
  o Storage-Network → dedicated for NFS datastore traffic.
  o vMotion-Network → dedicated for VM migration traffic.
• For each port group, a VM-kernel NIC was created:
o The Storage-Network VM-kernel NIC used Default TCP/IP stack and was assigned an IP address in the storage subnet.
o The vMotion-Network VM-kernel NIC used the vMotion TCP/IP stack and was assigned an IP address in the vMotion subnet.

Meanwhile, vSwitch0 continued to handle Management and VM traffic, with port groups for the management network and VM network.

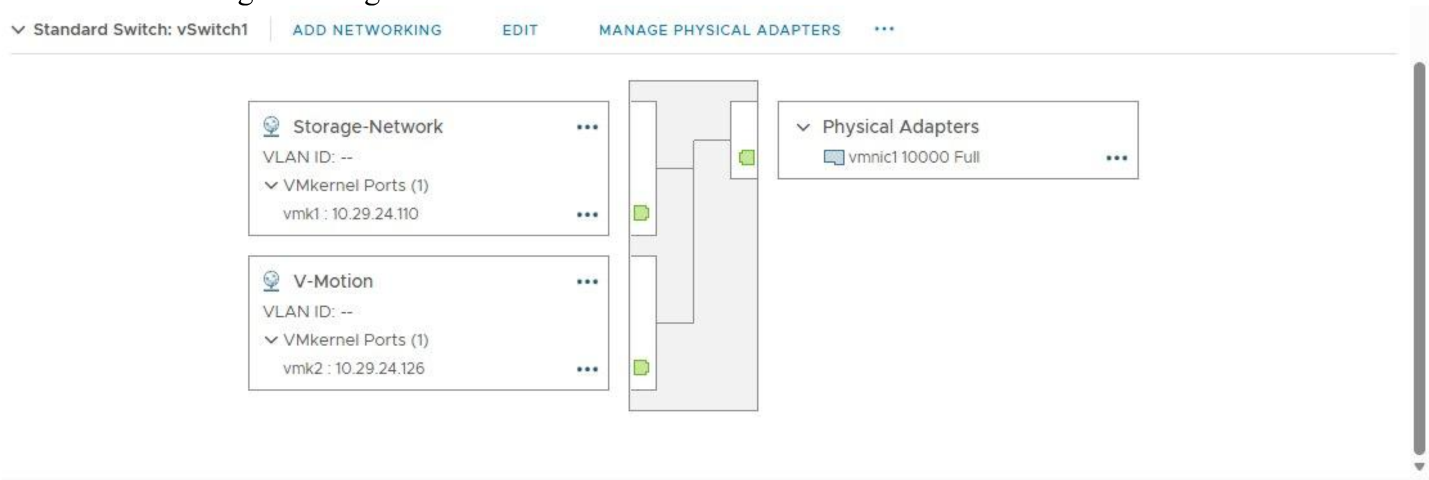
Why is it preferred to ensure Traffic Separation with Dedicated NICs?

To ensure reliable performance and proper isolation of traffic types, a new physical NIC was installed on each ESXi host and bound exclusively to **vSwitch1**. This design allowed storage and vMotion traffic to run on their own dedicated uplink, separate from management and VM traffic on vSwitch0. By assigning a physical NIC to vSwitch1, bandwidth contention was reduced, and critical operations such as datastore access and live migration were guaranteed consistent throughput. This approach follows VMware best practices for traffic separation and enhances both stability and efficiency in the cluster.

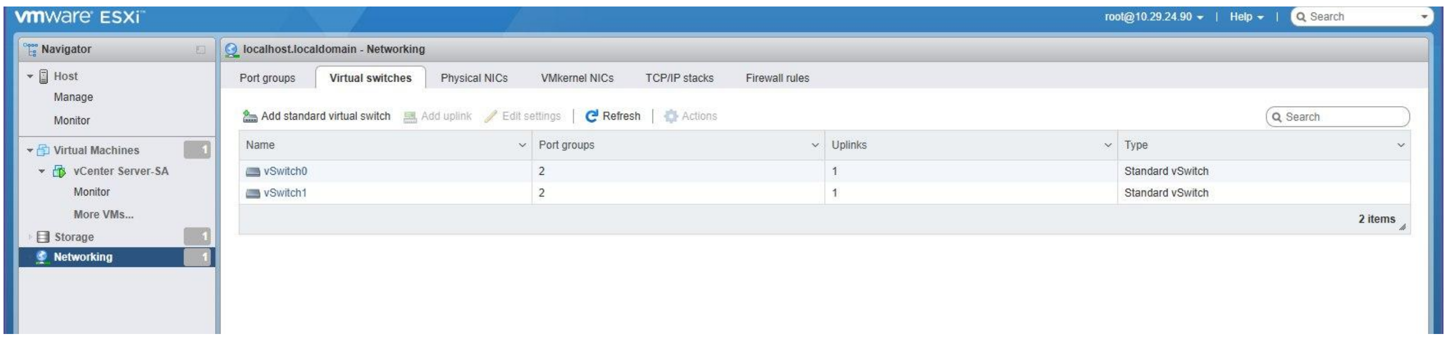
vSwitch0 for management and VMs port groups:



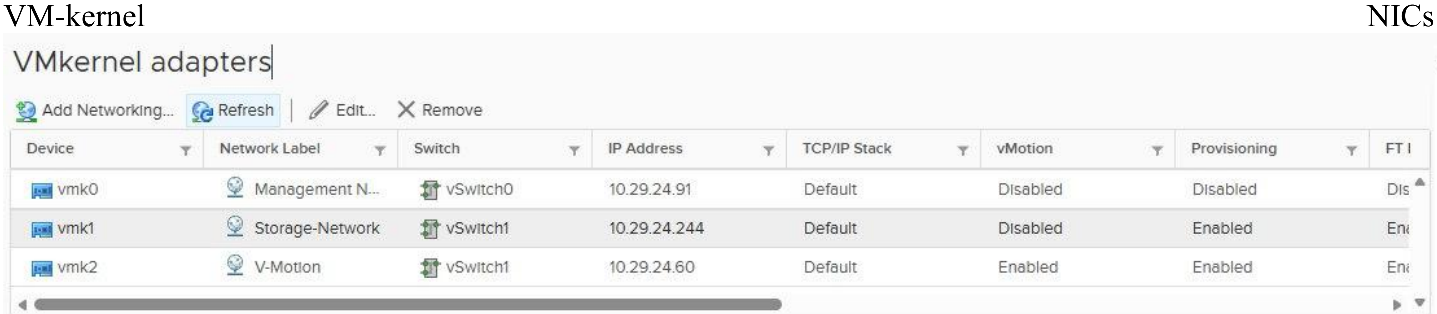
vSwitch1 for storage and migration:

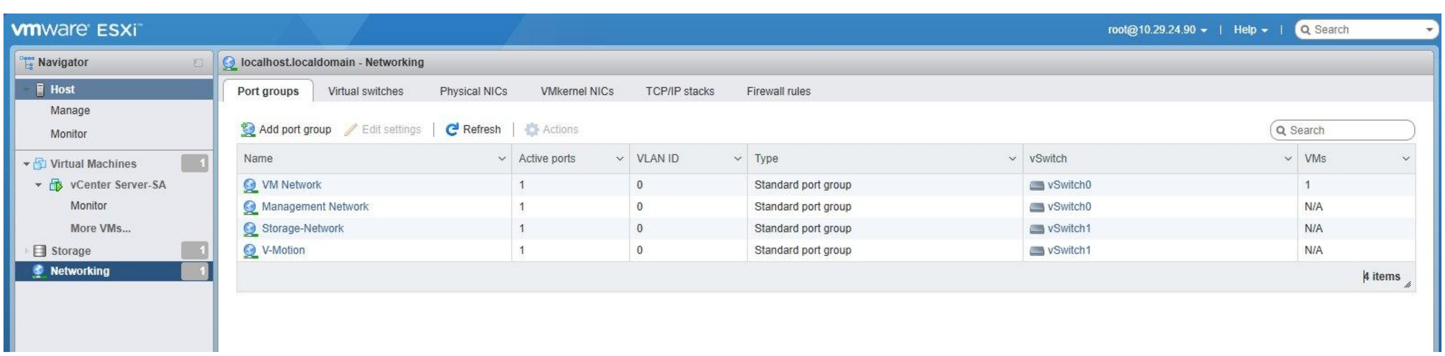


vSwitches with the ports:



VM-kernel NICs:



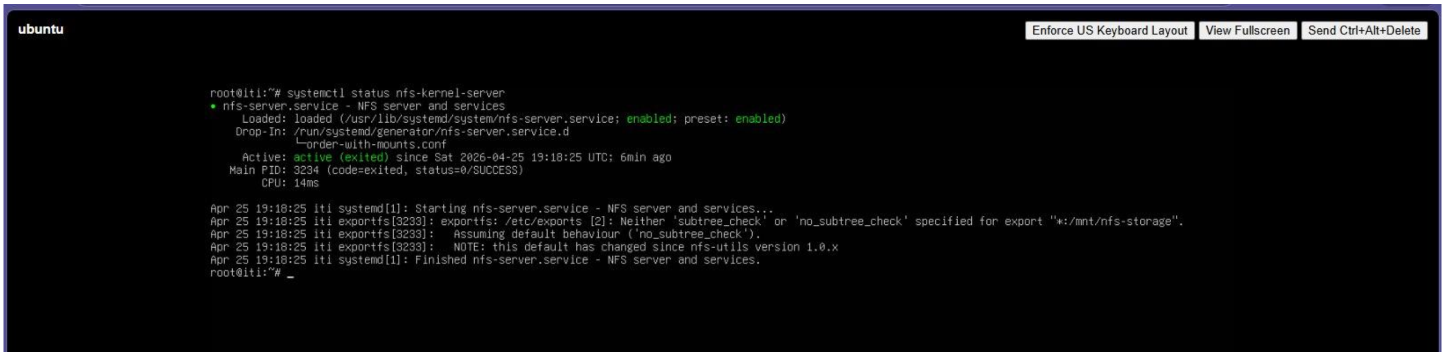


6. Shared **NFS Shared Storage Configuration**

6.1 NFS Server Setup

A dedicated Linux server was configured to provide centralized NFS This shared storage enables features like vMotion, High Availability, and Distributed Resource Scheduler.

6.1.1 Installing NFS Utilities and Service Status:



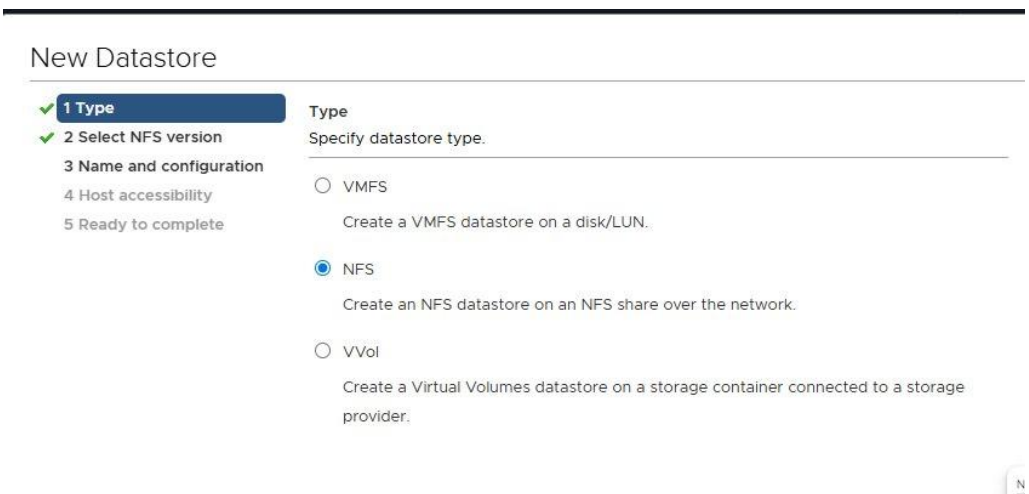
6.1.2

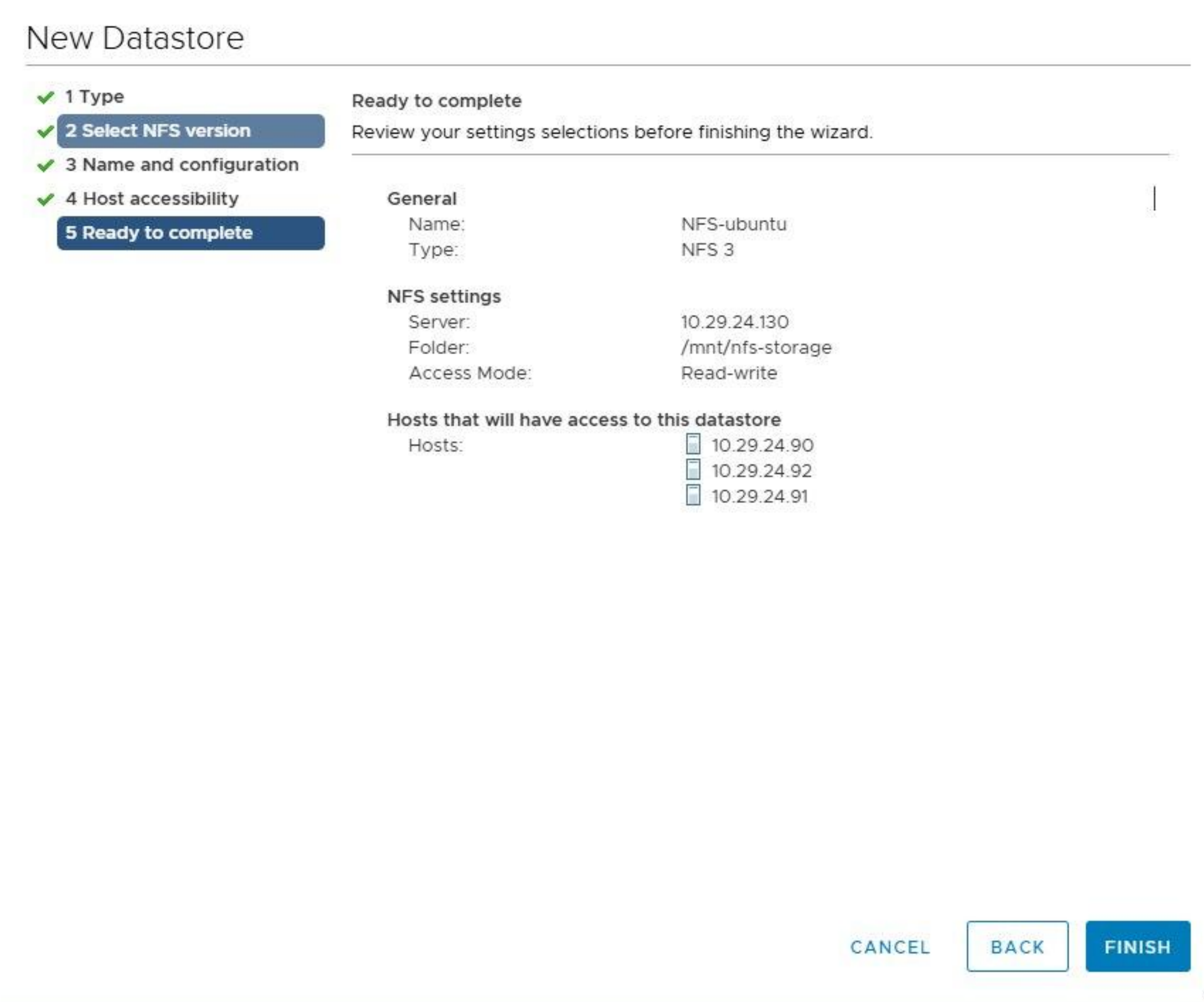
Creating and Mounting Storage Partition and NFS Export Configuration and Firewall Setup:

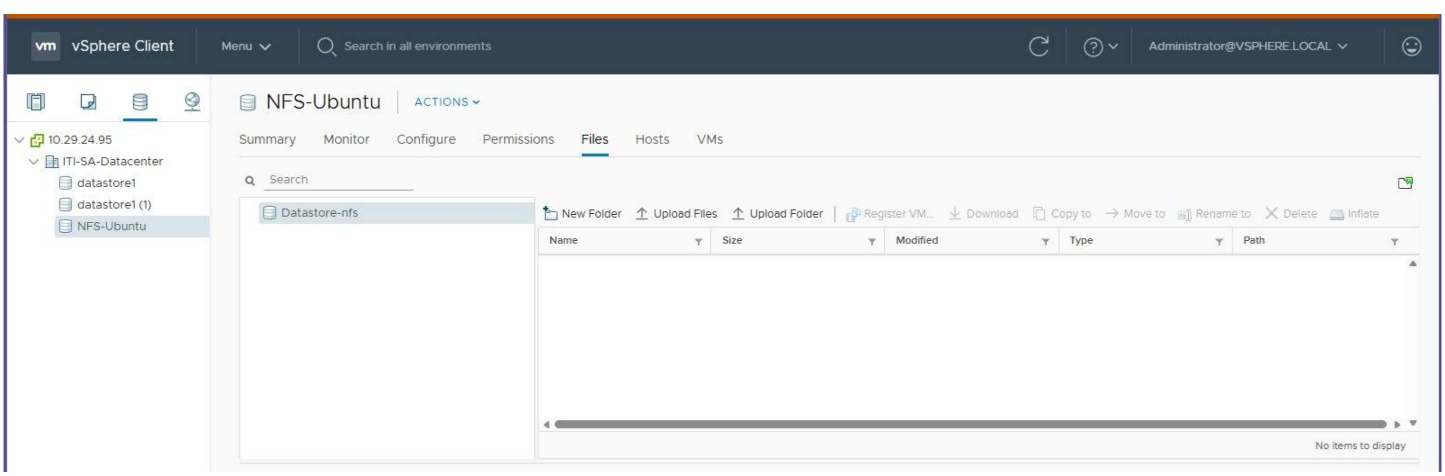
6.2

Adding NFS Datastore to vCenter

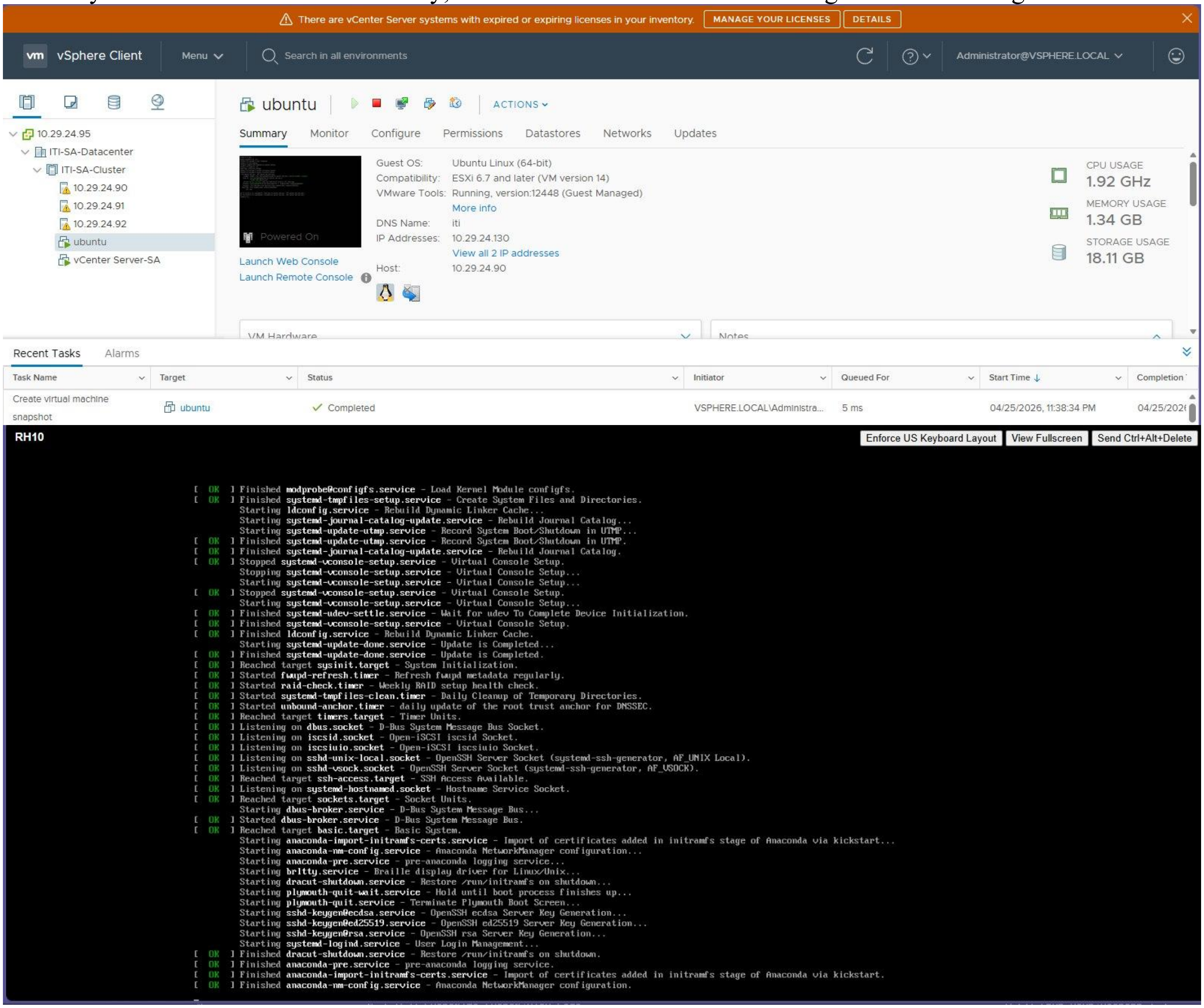
After configuring the NFS server, the shared storage was added as a datastore in vCenter




New Datastore
1 Type
2 Select NFS version
3 Name and configuration
4 Host accessibility
5 Ready to complete
Ready to complete
Review your settings selections before finishing the wizard.
General
Name: NFS-ubuntu
Type: NFS 3
NFS settings
Server: 10.29.24.130
Folder: /mnt/nfs-storage
Access Mode: Read-write
Hosts that will have access to this datastore
Hosts: 10.29.24.90
10.29.24.92
10.29.24.91
CANCEL
BACK
FINISH


vm vSphere Client
Menu
Search in all environments
Administrator@VSPHERE.LOCAL
10.29.24.95
ITI-SA-Datacenter
datastore1
datastore1 (1)
NFS-Ubuntu
NFS-Ubuntu
ACTIONS
Summary
Monitor
Configure
Permissions
Files
Hosts
VMs
Search
Datastore-nfs
New Folder
Upload Files
Upload Folder
Register VM...
Download
Copy to
Move to
Rename to
Delete
Inflate
Name
Size
Modified
Type
Path
No items to display

6.3 **Virtual Machine Deployment on NFS Storage**
To verify the NFS datastore functionality, a virtual machine was created using the shared storage.



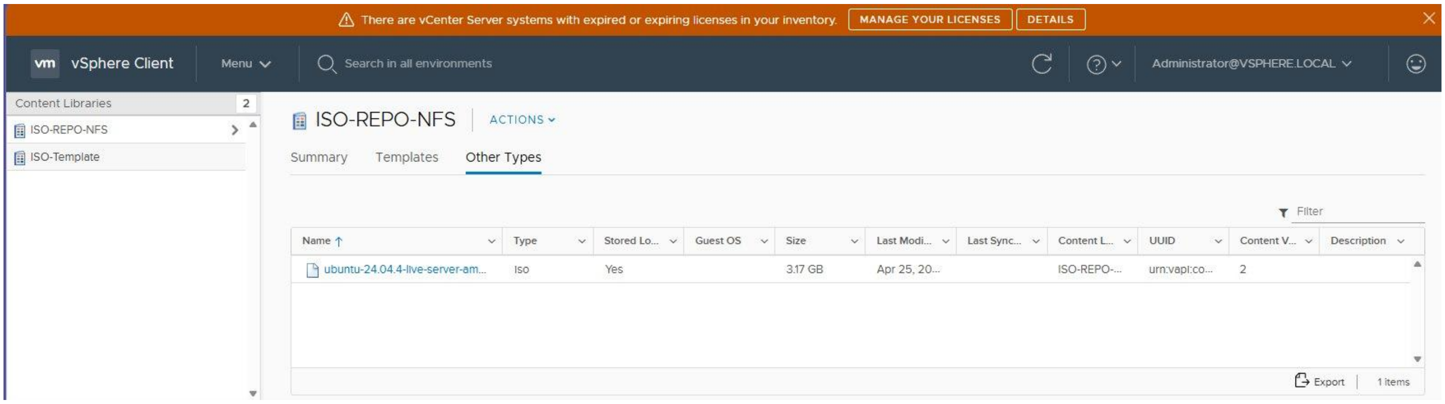
7 Content Library

7.1 Creating Content Library

After the NFS datastore was visible in vCenter, a **Content Library** was created to centralize ISO files for virtual machine deployment. The library was configured as a **local content library** and stored on the NFS data store to ensure accessibility by all ESXi hosts.

ISO files (Windows/Linux) were then uploaded into the content library. Once uploaded, the files appeared in the library items list, confirming that they were stored on the NFS datastore. This setup allows all hosts to use the same ISO images when creating new VMs, simplifying deployment and ensuring consistency across the environment.

7.1.1 Content Library Verification:



Naming Content Library: ISO-REPO-NFS
Configuring Library Type : local
Selecting Storage Location: NFS-datastore

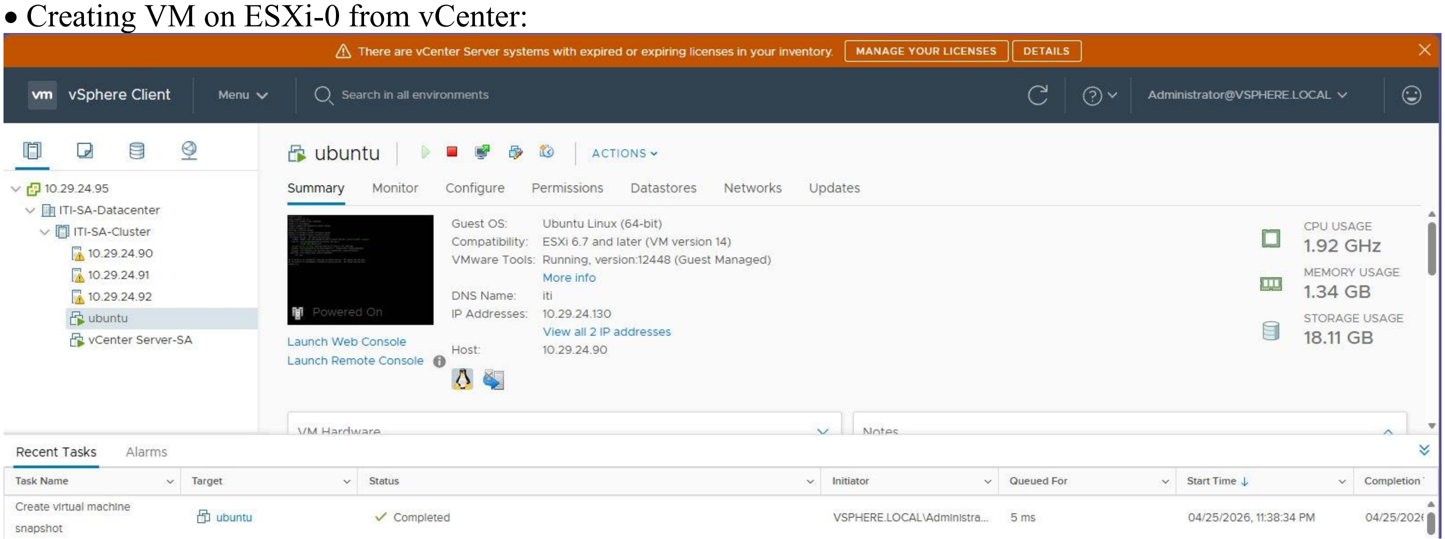
8. Virtual machine deployment:

With the NFS datastore and content library prepared, a new virtual machine was deployed through vCenter. The VM was created on the cluster and stored on the shared NFS datastore to ensure accessibility across all ESXi hosts.

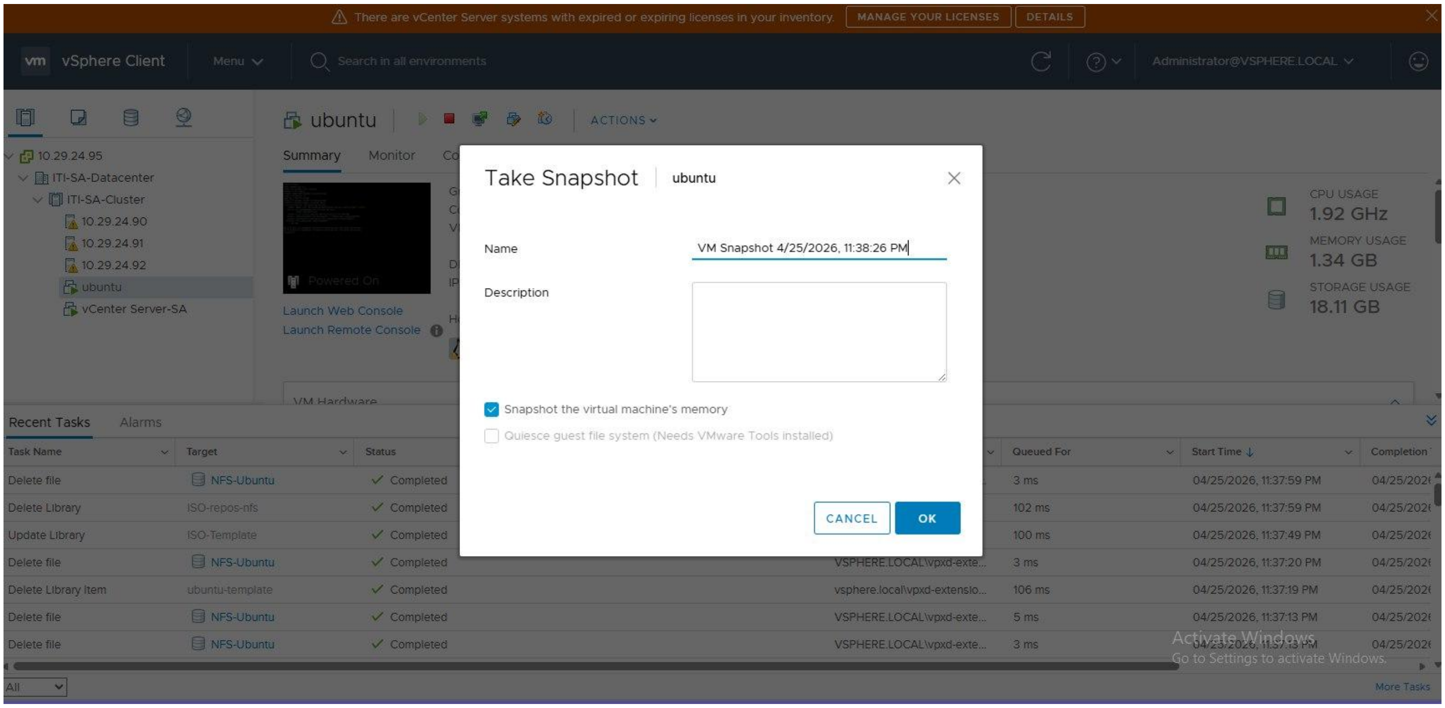
During configuration, CPU, memory, and disk resources were allocated, and the VM was connected to the VM_Network port group for communication. For installation media, the ISO file previously uploaded to the content library was attached to the VM's CD/DVD drive, allowing the operating system installation to begin directly from the centralized storage.

After powering on the VM, the OS installation was completed successfully, and the machine was tested for functionality and connectivity, confirming that the deployment process was effective and that the shared storage and content library were working as intended.

- Creating VM on ESXi-0 from vCenter:



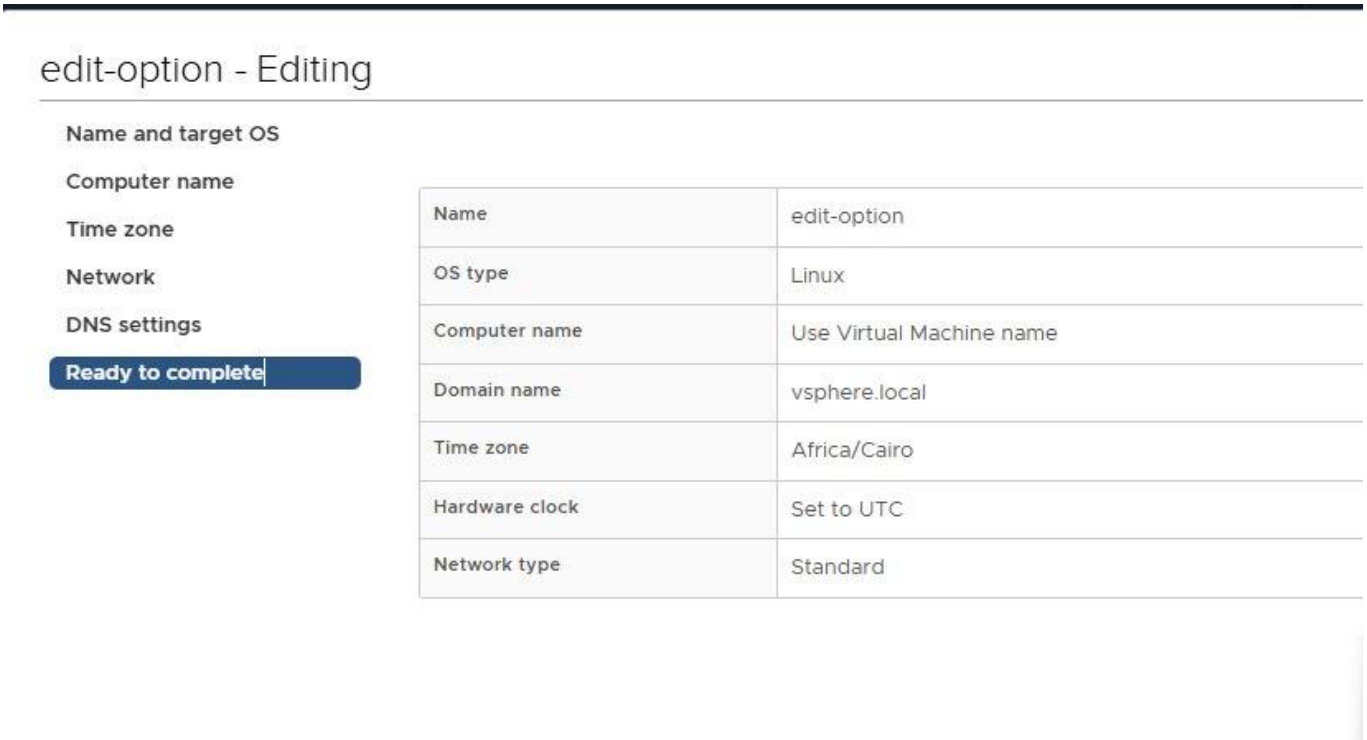
9.VM Operations
9.1 Snapshot
One of the key operations performed on virtual machines is the creation of **snapshots**. A snapshot captures the exact state of a VM at a specific point in time, including its disk, memory, and settings.
In vCenter, we selected the deployed VM and used the **Take Snapshot** option to record its current configuration before making changes. This feature is especially useful for testing and troubleshooting, as it allows the VM to be reverted back to the saved state if needed. After creating the snapshot, we verified its presence under the VM's Snapshot Manager, confirming that the operation was successful and ready to be used for rollback or recovery.
• VM snapshot:



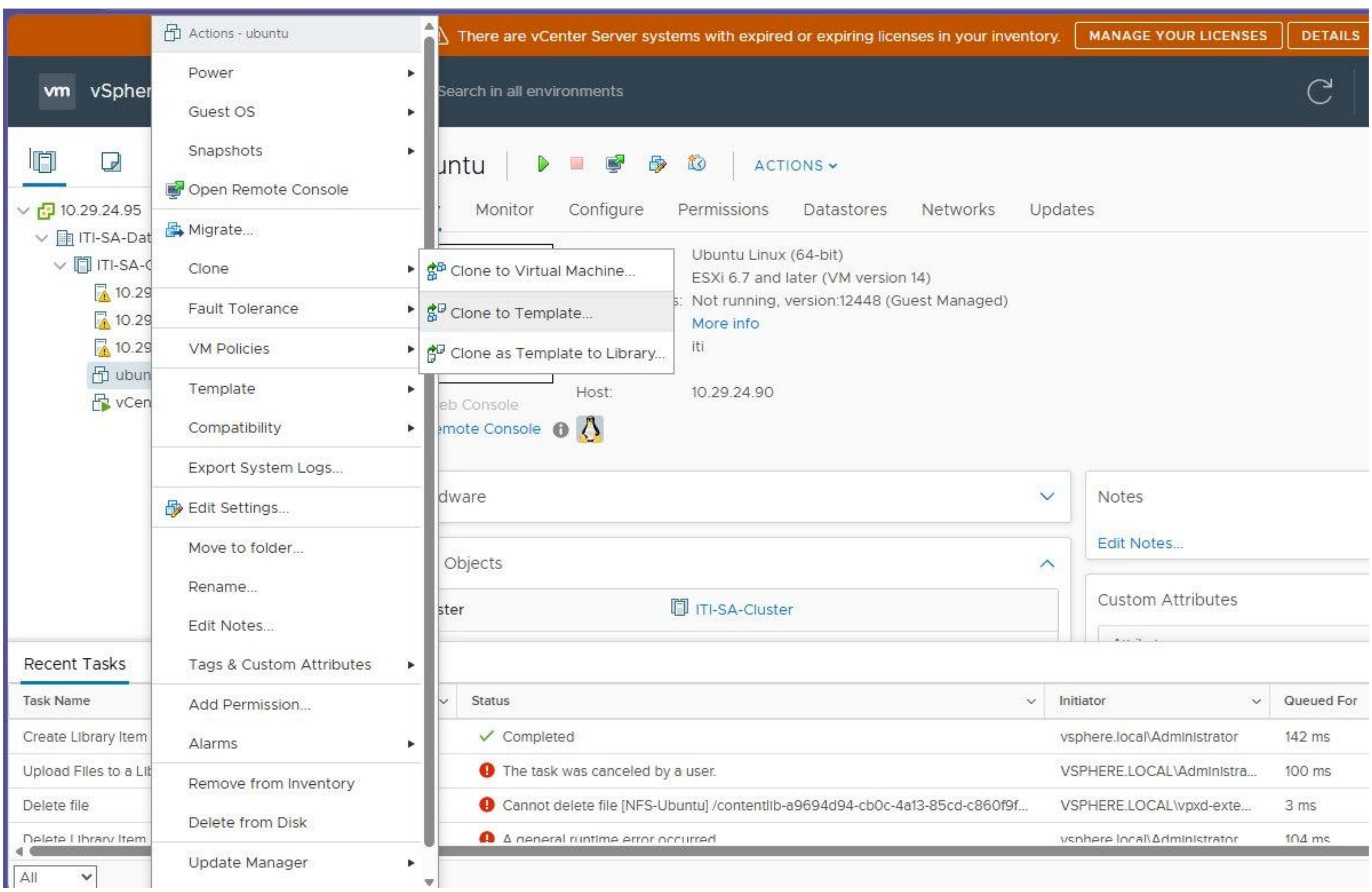
9.2 Clone to template

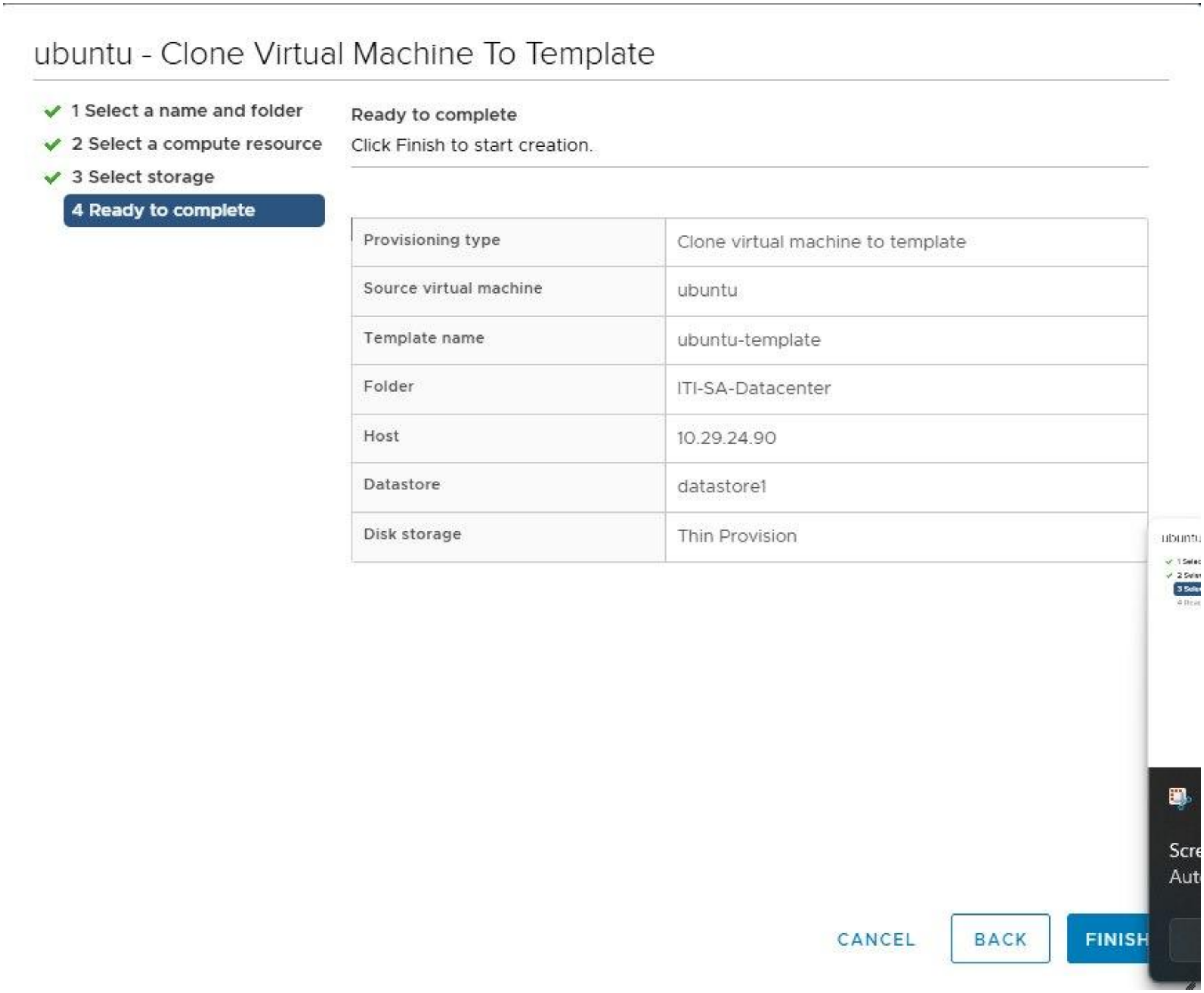
Another important operation performed on virtual machines is cloning and converting them into templates. In vCenter, we selected the deployed VM and used the **Clone to Template** option, which allows the VM's configuration and operating system to be preserved for future deployments.
During this process, a **VM Customization Specification** was required to define parameters such as computer name, network settings, and licensing details. This ensures that each new VM created from the template can be customized automatically without manual reconfiguration. Once the template was created, it appeared in the vCenter inventory and could be used to deploy multiple identical VMs quickly and consistently, streamlining the provisioning process across the cluster.

• Customization specification:

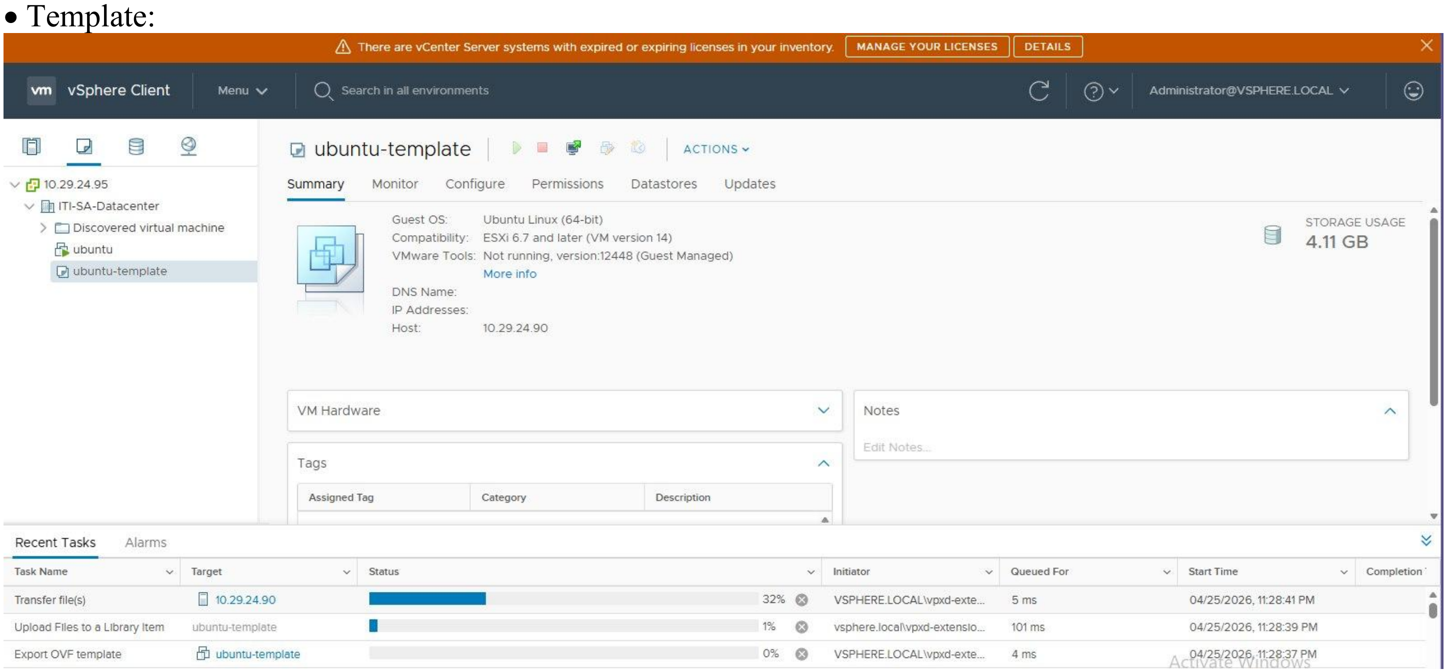


Clone to Template:




ubuntu - Clone Virtual Machine To Template
1 Select a name and folder
2 Select a compute resource
3 Select storage
4 Ready to complete
Ready to complete
Click Finish to start creation.
Provisioning type
Clone virtual machine to template
Source virtual machine
ubuntu
Template name
ubuntu-template
Folder
ITI-SA-Datacenter
Host
10.29.24.90
Datastore
datastore1
Disk storage
Thin Provision
CANCEL
BACK
FINISH

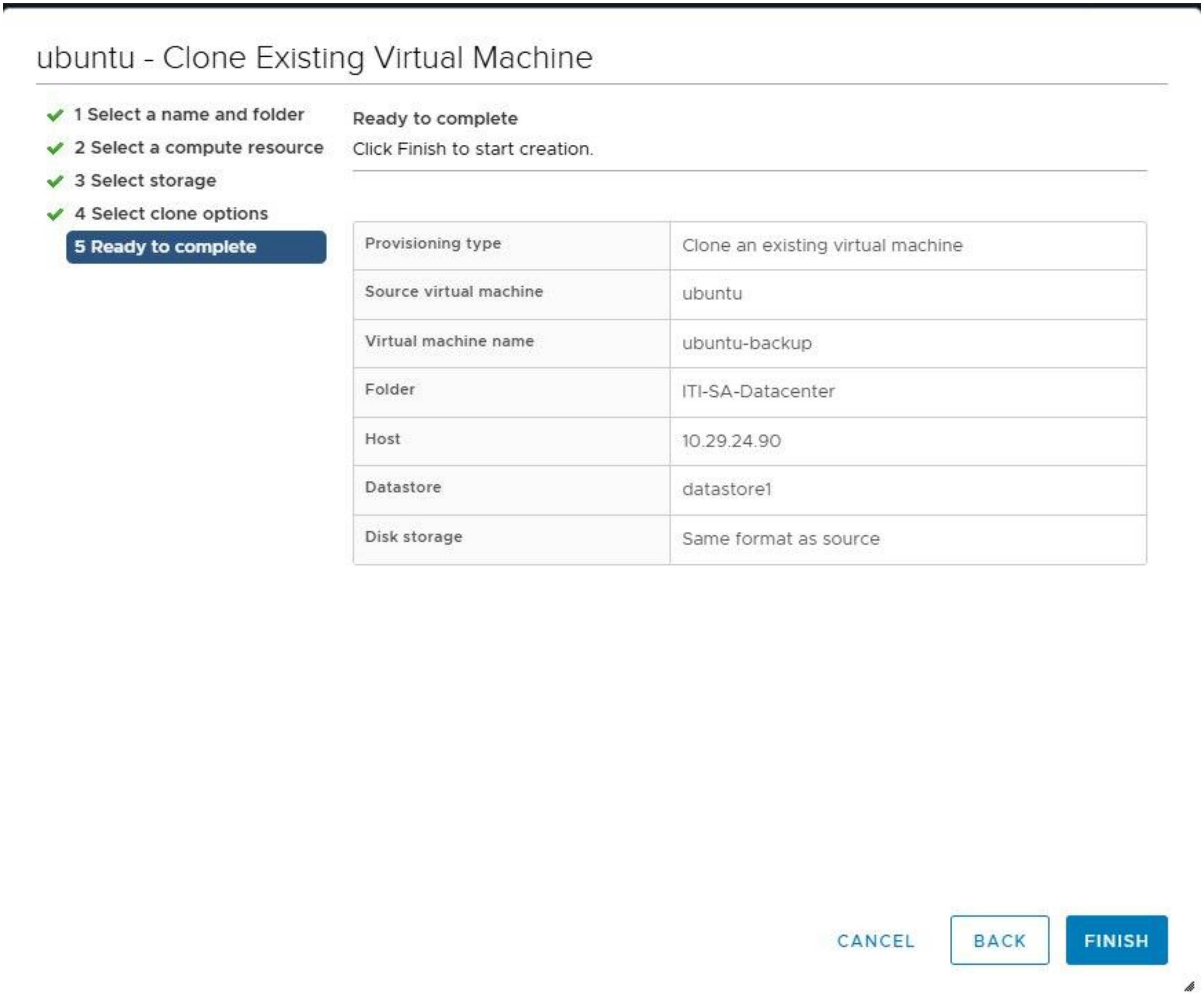
- Template:


There are vCenter Server systems with expired or expiring licenses in your inventory.
MANAGE YOUR LICENSES
DETAILS
vm vSphere Client
Menu
Search in all environments
Administrator@VSPHERE.LOCAL
10.29.24.95
ITI-SA-Datacenter
Discovered virtual machine
ubuntu
ubuntu-template
ubuntu-template
ACTIONS
Summary
Monitor
Configure
Permissions
Datastores
Updates
Guest OS: Ubuntu Linux (64-bit)
Compatibility: ESXi 6.7 and later (VM version 14)
VMware Tools: Not running, version:12448 (Guest Managed)
More info
DNS Name:
IP Addresses:
Host: 10.29.24.90
STORAGE USAGE
4.11 GB
VM Hardware
Notes
Edit Notes...
Tags
Assigned Tag
Category
Description
Recent Tasks
Alarms
Task Name
Target
Status
Initiator
Queued For
Start Time
Completion
Transfer file(s)
10.29.24.90
32%
VSPHERE.LOCAL\vpxd-exte...
5 ms
04/25/2026, 11:28:41 PM
Upload Files to a Library Item
ubuntu-template
1%
vsphere.local\vpxd-extensio...
101 ms
04/25/2026, 11:28:39 PM
Export OVF template
ubuntu-template
0%
VSPHERE.LOCAL\vpxd-exte...
4 ms
04/25/2026, 11:28:37 PM
Activate Windows

9.3 Clone

Cloning a virtual machine provides an exact copy of an existing VM, including its operating system, applications, and configuration. In vCenter, we selected the deployed VM and initiated the **Clone** operation. During the process, the destination ESXi host and datastore were specified, ensuring that the cloned VM was placed directly on one of the available ESXi hosts within the cluster.

• Clone:



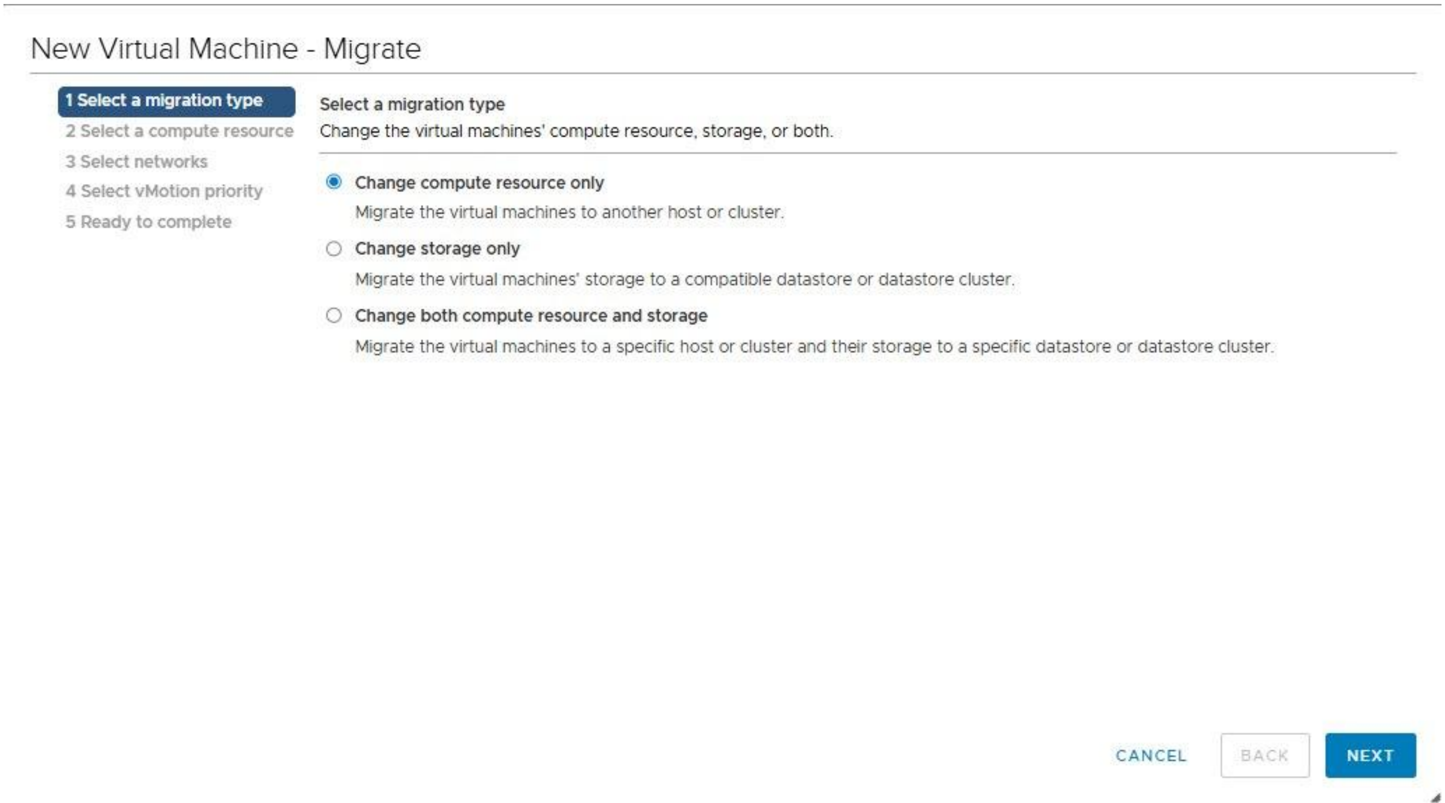
10. vMotion

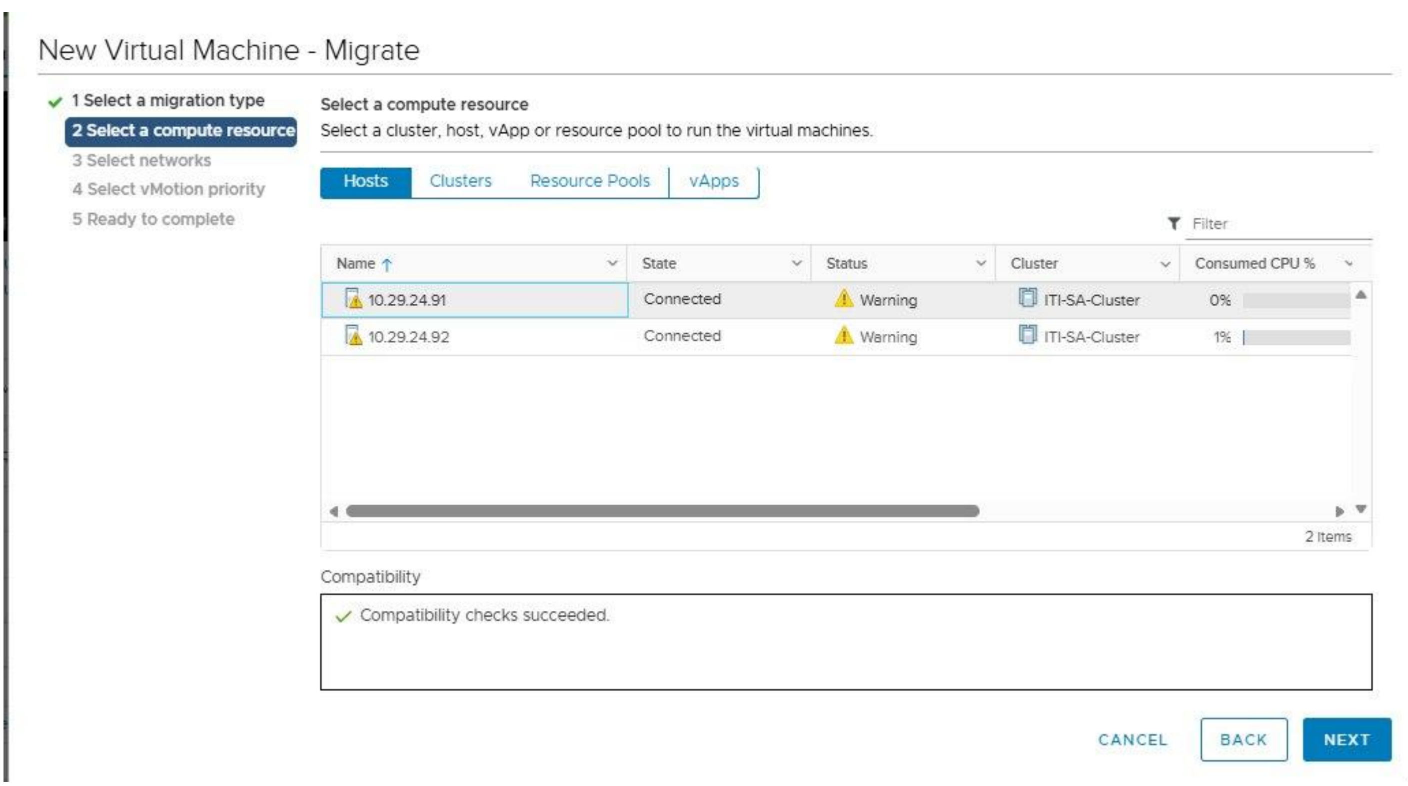
VMware vMotion was configured to enable the live migration of virtual machines between ESXi hosts without downtime. The process began by creating a new physical NIC on each host, which was bound exclusively to vSwitch1. On this switch, a dedicated port group named vMotion-Network was created, and a VM-kernel adapter was assigned with an IP address in the vMotion subnet. This adapter implicitly carried the vMotion service, ensuring that migration traffic was isolated from management and VM traffic on vSwitch0.

Once the vMotion network was in place, a test migration was performed: a VM running on ESXi1 was migrated to ESXi0 using the “Change compute resource only” option in the vSphere Client. The migration succeeded because both hosts shared access to the NFS datastore and communicated over the dedicated vMotion network on vSwitch1.

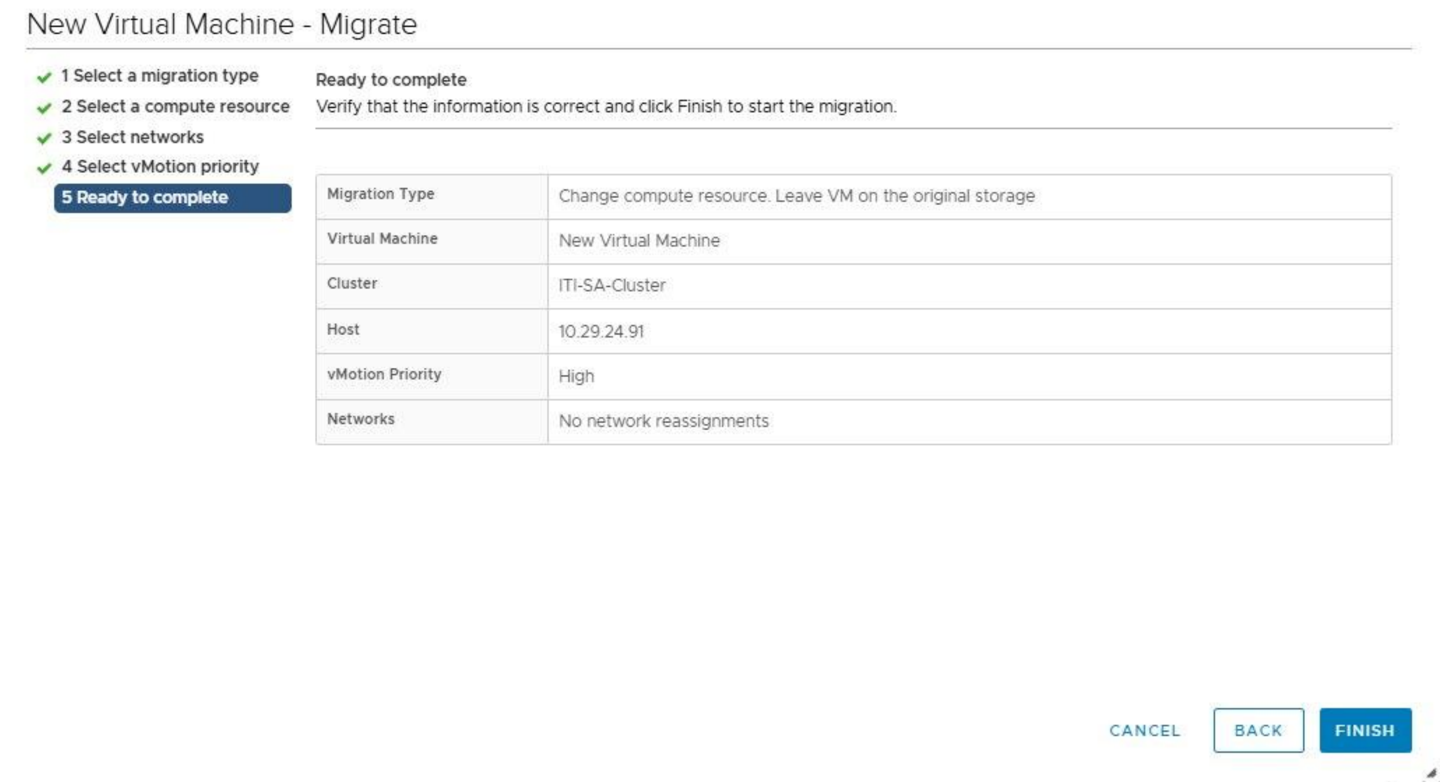
This setup demonstrated the importance of traffic separation: by using a new physical NIC for vSwitch1, vMotion traffic was given its own bandwidth and path, preventing congestion with other services and ensuring smooth, reliable live migrations across the cluster.

- VM was on ESXi-2:





- Migration:



- VM migrated to ESXi-1:

11. . High availability (HA)

VMware High Availability (HA) was configured to protect virtual machines against host failures. In this setup, all ESXi hosts were added to the same cluster and connected to a shared NFS datastore. The shared datastore is critical because it allows any surviving host to access the VM's files and restart them if another host goes down. HA continuously monitors host health, and when a failure is detected, affected VMs are automatically restarted on another available host in the cluster.

By combining HA with the dedicated networking design—management and VM traffic on vSwitch0, and storage plus vMotion traffic on vSwitch1—the environment ensures both resilience and performance. The use of vSwitch1 with its dedicated NIC guarantees that storage and vMotion traffic remain available during failover, enabling HA to function reliably.

• Configuration:

• After simulating Host ESXi-2 failure all VMs now working on ESXi-1:

At HA, the VM took time to restart.

12. Distributed Resource Scheduler (DRS)

VMware Distributed Resource Scheduler (DRS) was enabled to balance workloads across ESXi hosts in the cluster. DRS relies on vMotion to migrate virtual machines automatically between hosts, ensuring optimal use of CPU and memory resources. In this project, the configuration of a dedicated NIC on vSwitch1 for vMotion traffic was critical, as it provided the bandwidth and isolation needed for seamless migrations. With shared access to the NFS datastore, DRS can move VMs without downtime, preventing resource contention and maintaining performance across the cluster.

• Configuration:

Hint: Aggressive was chosen just for ease in verification

To verify Distributed Resource Scheduler (DRS), a load imbalance was created by powering on multiple VMs on ESXi-1, increasing its CPU and memory usage. With DRS set to fully automated mode, the system detected the imbalance and initiated vMotion migrations to ESXi-0. The VMs were successfully redistributed across the hosts, confirming that DRS was operational. This test demonstrated the importance of the dedicated vMotion network on vSwitch1, which ensured smooth, automated workload balancing without downtime

13. Fault Tolerance (FT)
VMware Fault Tolerance (FT) was configured to provide continuous availability for critical virtual machines. Unlike High Availability, which restarts VMs after a host failure, FT creates a live shadow instance of the VM on another host. Both the primary and secondary VM run in lockstep, ensuring zero downtime and no data loss if the primary host fails.

In this project, FT relied on shared storage through the NFS datastore and the dedicated vMotion network on vSwitch1, which carried the synchronization traffic between hosts. This setup demonstrated the importance of traffic separation, as FT requires consistent bandwidth and low latency to keep the primary and secondary VMs synchronized.

A dedicated **FT network** was added to **vSwitch1**, using the new physical NIC to isolate FT traffic from other services. This separation ensured that the continuous synchronization between primary and secondary VMs had sufficient bandwidth and low latency, preventing interference with management, VM, or vMotion traffic. The use of vSwitch1 for FT traffic was critical to maintaining reliable performance and stability in the cluster.

- Dedicated port:

To verify VMware Fault Tolerance (FT), a VM named **ARY-VM** was initially running on **ESXi-1** with FT enabled. The host **ESXi-1** was then powered off to simulate a failure. Immediately, the secondary VM on **ESXi-0** took over, and ARY-VM continued running without interruption. This confirmed that FT was operational, providing zero downtime and seamless failover between hosts.

- **VM** was initially running on **ESXi-1** with FT enabled:

- After ESXi-1 is powered off, the VM immediately continue running on

ESXi-0. No downtime, No restarting as what happened in HA:

14.Conclusion:

This project provided a comprehensive, hands-on exploration of VMware vSphere 6.7 within a nested virtualization environment built entirely on two physical laptops using VMware Workstation. By designing and deploying a functional private cloud infrastructure from scratch, the team gained practical experience across a wide range of enterprise virtualization concepts that are directly applicable to real-world IT environments.

Infrastructure Deployment

The foundation of the lab was successfully established by installing two ESXi 6.7 hosts as virtual machines within VMware Workstation and deploying the vCenter Server Appliance (VCSA) to centrally manage them. Connecting both hosts to vCenter and organizing them into a cluster enabled the advanced features required for the remaining tasks. This phase reinforced the importance of proper planning — particularly around network addressing and resource allocation — before any virtual infrastructure deployment begins.

Networking and Storage

A deliberate two-switch networking design was implemented to separate traffic types appropriately. vSwitch0 was dedicated to management and virtual machine traffic, while vSwitch1 — attached to a second virtual NIC — carried storage and vMotion traffic on an isolated subnet. This separation mirrors enterprise best practices and directly demonstrates the performance and reliability benefits of dedicated network paths. The NFS datastore was successfully configured and mounted on both hosts, providing shared storage essential for advanced cluster features such as vMotion and High Availability.

Virtual Machine Operations

A guest virtual machine was created, and an operating system was installed to validate the environment end-to-end. The team then performed core VM lifecycle operations including snapshot creation, cloning, and template conversion. These operations are fundamental to enterprise VM management workflows and demonstrate how vSphere enables rapid provisioning and consistent deployments through reusable templates and content libraries.

High Availability, DRS, and Fault Tolerance

The most critical validation of the lab environment was the High Availability failover test. By simulating a host failure — suspending the ESXi Host 2 VM in Workstation — it was confirmed that vCenter detected the failure and automatically restarted the affected virtual machine on ESXi Host 1 within the expected timeout window. This demonstrated that HA was correctly configured with admission control policies in place.

Distributed Resource Scheduler (DRS) was configured in Fully Automated mode, allowing vCenter to balance virtual machine workloads across hosts without manual intervention. Finally, Fault Tolerance was enabled on a selected VM, providing continuous availability through a synchronized secondary VM on the alternate host — a level of protection beyond what HA alone can offer.

Challenges and Lessons Learned

Several practical challenges were encountered and resolved during the project:

- Nested virtualization required enabling hardware virtualization flags (VT-x/EPT) on each ESXi VM in Workstation — a non-obvious step that is critical for running guest VMs inside ESXi.

- The two-switch networking requirement initially appeared to demand a second physical NIC. This was resolved by adding a second virtual network adapter to each ESXi VM in Workstation, mapped to VMnet2, allowing vSwitch1 to be created without any additional physical hardware.

- vMotion across two physical laptops required careful alignment of VMnet subnets on both machines so that the vMotion VM-kernel adapters could communicate across the physical LAN.

- Fault Tolerance imposed strict prerequisites — no snapshots, single vCPU, and a dedicated FT logging VM-kernel NIC — reinforcing that production FT deployments require careful planning.

Overall, the project successfully met all stated objectives. The experience of building this environment from the ground up — including troubleshooting real configuration issues — provided deeper understanding than any theoretical study could achieve. The skills developed throughout this project, including datacenter design, cluster management, shared storage configuration, and high availability planning, form a strong foundation for pursuing VMware certifications and working in professional virtualization and cloud infrastructure roles.

15.Acknowledgement

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